

# Thomas Weng

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Robotics Research Scientist focusing on robot post-training for contact-rich manipulation:  
finetuning pretrained policies for deployment scenarios through real-world interaction.

## EDUCATION

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### Carnegie Mellon University

Ph.D. in Robotics

Advisor: Dave Held

Thesis Committee: Dave Held, Oliver Kroemer, Shubham Tulsiani, Alberto Rodriguez (MIT)

2018 - 2023

Pittsburgh, PA

### Yale University

B.S. Computer Science & B.A. Economics

GPA: 3.77 / 4.0 with distinction in the C.S. major

Senior Thesis Advisor: Brian Scassellati

2011 - 2015

New Haven, CT

## PUBLICATIONS

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- [C12] Dodeja, L., Biza, O., Vats, S., Hart, S., Tellex, S., Walters, R., Schmeckpeper, K., and **Weng, T.** When Life Gives You BC, Make Q-functions: Extracting Q-values from Behavior Cloning for On-Robot Reinforcement Learning. *Robotics: Science and Systems (RSS)*, 2026.
- [C11] Schroeder, P., Biza, O., **Weng, T.**, Luo, H., Glass, J. ROVER: Recursive Reasoning Over Videos with Vision-Language Models for Embodied Tasks. *Conference on Neural Information Processing Systems (NeurIPS)*, 2025.
- [C10] Dodeja, L., Schmeckpeper, K., Vats, S., **Weng, T.**, Jia, M., Konidaris, G., Tellex, S. Accelerating Residual Reinforcement Learning with Uncertainty Estimation. *IEEE Robotics and Automation Letters*, 2025. The contents of this paper were also selected by the ICRA 2026 Program Committee for presentation at the conference.
- [C9] Biza, O., **Weng, T.**, Sun, L., Schmeckpeper, K., Kelestemur, T., . . . Wong, L. L. On-Robot Reinforcement Learning with Goal-Contrastive Rewards. In *IEEE International Conference on Robotics and Automation (ICRA)*, 2025.
- [C8] **Weng, T.**, Held, D., Meier, F., and Mukadam, M. Neural Grasp Distance Fields for Robot Manipulation. *IEEE International Conference on Robotics and Automation (ICRA)*, 2023.
- [C7] Tirumala, S.\*, **Weng, T.\***, Seita, D.\*, Kroemer, O., Temel, Z., and Held, D. Learning to Singulate Layers of Cloth Based on Tactile Feedback. *IEEE International Conference on Intelligent Robots and Systems (IROS)*, 2022.
- [C6] **Weng, T.**, Bajracharya, S., Wang, Y., Agrawal, K. and Held, D. FabricFlowNet: Bimanual Cloth Manipulation with a Flow-based Policy. *Conference on Robot Learning (CoRL)*, 2021.
- [C5] Qian, J.\*, **Weng, T.\***, Okorn, B., Zhang, L., and Held, D. Cloth Region Segmentation for Robust Grasp Selection. *International Conference on Intelligent Robots and Systems (IROS)*, 2020.
- [J1] **Weng, T.**, Pallankize, A., Tang, Y., Kroemer, O., and Held, D. Multi-modal perception and transfer learning for grasping transparent and specular objects. *IEEE Robotics and Automation Letters*, 2020. The contents of this paper were also selected by the ICRA 2020 Program Committee for presentation at the conference.
- [C4] **Weng, T.**, Perlmutter, L., Nikolaidis, S. , Srinivasa, S., and Cakmak, M. Robot Object Referencing through Situated Legible Projections. *IEEE International Conference on Robotics and Automation (ICRA)*, 2019.
- [C3] Sefidgar, Y.\*, **Weng, T.\***, and Cakmak, M. RobotIST: Interactive Situated Tangible Robot Programming. *Proceedings of the Symposium on Spatial User Interaction*. ACM, 2018.
- [C2] Admoni, H., **Weng, T.**, and Scassellati, B. Modeling communicative behaviors for object references in human-robot interaction. *IEEE International Conference on Robotics and Automation (ICRA)*, 2016.
- [C1] Admoni, H., **Weng, T.**, Hayes, B. and Scassellati, B. Robot nonverbal behavior improves task performance in difficult collaborations. *ACM/IEEE International Conference on Human Robot Interaction (HRI)*, 2016.

## RESEARCH AND INDUSTRY EXPERIENCE

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|                                                                                                                                                                                                                                      |               |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|
| <b>Robotics and AI Institute</b><br>Robotics Research Scientist<br><i>Researching sample-efficient skill learning for manipulation [C9-12].</i>                                                                                      | 2023 -present |
| <b>Facebook Research and AI Mentorship Program</b><br>Visiting Researcher with Dr. Mustafa Mukadam<br><i>Published on neural field representations for manipulation through a collaborative CMU-Meta program [C8].</i>               | 2021 -2023    |
| <b>University of Washington Human-Centered Robotics Lab</b><br>Research Scientist with Prof. Maya Cakmak<br><i>Published papers on tangible robot programming and robot-mounted projectors for human-robot interaction [C3, C4].</i> | 2017 - 2018   |
| <b>Microsoft Corp., AI and Research</b><br>Software Engineer on Bing<br><i>Worked on Bing Answers for enterprise Q&amp;A, flight booking, and the 2016 presidential election.</i>                                                    | 2015 - 2017   |
| <b>Yale University Social Robotics Lab</b><br>Undergraduate Researcher with Prof. Brian Scassellati<br><i>Published papers on modeling and generating robot non-verbal gestures [C1, C2].</i>                                        | 2014 - 2015   |

## AWARDS AND HONORS

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|-------------------------------------------------------------------------------------|------------|
| <b>Graduate Research Fellowship Award</b> , National Science Foundation             | 2019       |
| <b>Graduate Research Fellowship Honorable Mention</b> , National Science Foundation | 2018       |
| <b>Computer Science Research Prize</b> , Yale University                            | 2015       |
| <b>Trumbull College Scholarship for Economics</b> , Yale University                 | 2014       |
| <b>Maher Family Scholarship</b> , Yale University                                   | 2013, 2014 |

## MENTORSHIP AND TEACHING

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|                                                                                 |                |
|---------------------------------------------------------------------------------|----------------|
| <b>Lakshita Dodeja</b> , RAI Intern                                             | 2025 - present |
| <b>James Staley</b> , RAI Intern                                                | 2025           |
| <b>Mansi Agrawal</b> , M.S. Robotics                                            | 2021 - 2023    |
| <b>Sashank Tirumala</b> , M.S. Robotics                                         | 2021 - 2023    |
| <b>Sujay Bajracharya</b> , M.S. Robotics                                        | 2020 - 2021    |
| <b>Rashmi Anil</b> , CMU undergraduate                                          | 2019 - 2020    |
| <b>Khush Agrawal</b> , RI Summer Scholar                                        | Summer 2020    |
| <b>Yimin Tang</b> , RI Summer Scholar                                           | Summer 2019    |
| <b>Amith Pallankize</b> , visiting undergraduate                                | 2018 - 2019    |
| <b>Teaching Assistant</b> , CMU 16-720 Computer Vision                          | Spring 2021    |
| <b>Teaching Assistant</b> , CMU 16-811 Math Fundamentals for Robotics           | Fall 2020      |
| <b>Mentor</b> , CMU Masters in Research and Software Development (MRSD) Team    | 2018 - 2019    |
| <b>Teaching Assistant</b> , University of Washington CSE 481C Robotics Capstone | Spring 2017    |

## SERVICE AND LEADERSHIP EXPERIENCE

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### Reviewer

Robotics: Science and Systems (RSS)  
Conference on Robot Learning (CoRL)  
Robotics and Automation Letters (RA-L)  
International Conference on Robotics and Automation (ICRA)  
International Conference on Intelligent Robots and Systems (IROS)  
Human Robot Interaction (HRI)  
Humanoids  
Transactions on Mechatronics  
NeurIPS Workshop: Black in AI  
Masters in Computer Vision (MSCV) Admissions, 2020

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| <b>Graduate Student Assembly (GSA) Representative</b> , Carnegie Mellon University<br><i>Elected representative of graduate students at the CMU Robotics Institute.</i>                                                                                               | 2018 - 2023 |
| <b>Legislative Action Days Delegate</b><br><i>Met with senatorial and congressional staff in Washington, D.C. to advocate on behalf of graduate students on policy issues such as research funding, immigration reform, food insecurity, mental health, and more.</i> | 2019, 2021  |

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|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------|
| <b>AI Undergraduate Research Mentor, CMU</b><br><i>Met monthly with an undergraduate from an underrepresented background on conducting research at CMU.</i>                                                                                              | <b>2019</b>        |
| <b>Girls of Steel FIRST Robotics Team Mentor, CMU Robotics Institute</b><br><i>Mentored the all-girls, multi-school team for the FIRST Robotics competition.</i>                                                                                         | <b>2018</b>        |
| <b>Tour Manager, Yale Alley Cats a cappella group</b><br><i>Managed tours for 14 undergraduates to the American South, the Pacific Northwest, Europe, and Asia.<br/>The Yale Alley Cats is one of the nation's most well-traveled a cappella groups.</i> | <b>2012 - 2014</b> |

## INVITED TALKS AND OUTREACH

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|                                                                                                                                                                      |                       |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------|
| <b>Learning Representations for Manipulating Everyday Objects</b><br>Robotics and AI Institute (formerly Boston Dynamics AI Institute)                               | <b>September 2023</b> |
| <b>Visual Action Representations for Contact-Centric Manipulation</b><br>University of Pennsylvania                                                                  | <b>September 2023</b> |
| New York University                                                                                                                                                  | <b>September 2023</b> |
| Columbia University                                                                                                                                                  | <b>August 2023</b>    |
| UC San Diego                                                                                                                                                         | <b>August 2023</b>    |
| <b>Google x Prep for Prep x VEX Robotics Workshop Guest Speaker, New York, NY</b><br><i>Presented on robotics to 100 gifted students of color in 6th-10th grade.</i> | <b>2022, 2023</b>     |
| <b>Buckley School Alumni Speaker, New York, NY</b><br><i>Gave presentations on robotics and tech career paths to 7th-9th graders.</i>                                | <b>2018, 2020</b>     |
| <b>Code Haven at Yale Guest Speaker, New Haven, CT</b><br><i>Spoke with students at under-served New Haven public schools about STEM careers.</i>                    | <b>2017</b>           |

## TECHNICAL SKILLS

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### Graduate Coursework

|        |                                                    |
|--------|----------------------------------------------------|
| 16-811 | Mathematical Foundations for Robotics              |
| 16-720 | Computer Vision                                    |
| 10-701 | Machine Learning                                   |
| 16-711 | Kinematics, Dynamics, and Control                  |
| 10-703 | Deep Reinforcement Learning and Control            |
| 16-782 | Planning and Decision-making in Robotics           |
| 10-725 | Convex Optimization                                |
| 16-881 | Deep Reinforcement Learning for Robotics (seminar) |

### Robotics

|                  |                                                               |
|------------------|---------------------------------------------------------------|
| Languages        | C/C++, Python, MATLAB                                         |
| Machine Learning | PyTorch, Tensorflow                                           |
| Tools            | ROS, MoveIt!, OpenCV, MuJoCo, PyBullet, Nvidia FleX, Unity    |
| Robots           | Franka Panda, Sawyer, PR2, Fetch, Baxter, Kuka, Aldebaran Nao |
| Sensors          | Azure Kinect DK, Kinect v2, Intel RealSense, Primesense       |

### Web Development

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|------------|---------------------------------------|
| Languages  | JavaScript, C#.NET, Python, Ruby      |
| Frameworks | React, Node.js, Django, Ruby on Rails |